

Roots of Equations

A simple introduction to $f(x)=0$, polynomials, and sign rules

$$f(x) = 0$$

Goal: find the value(s) of x

Answer may be exact or approximate

Used in science & engineering

What is a root?

A root is an input value that makes the function become zero.

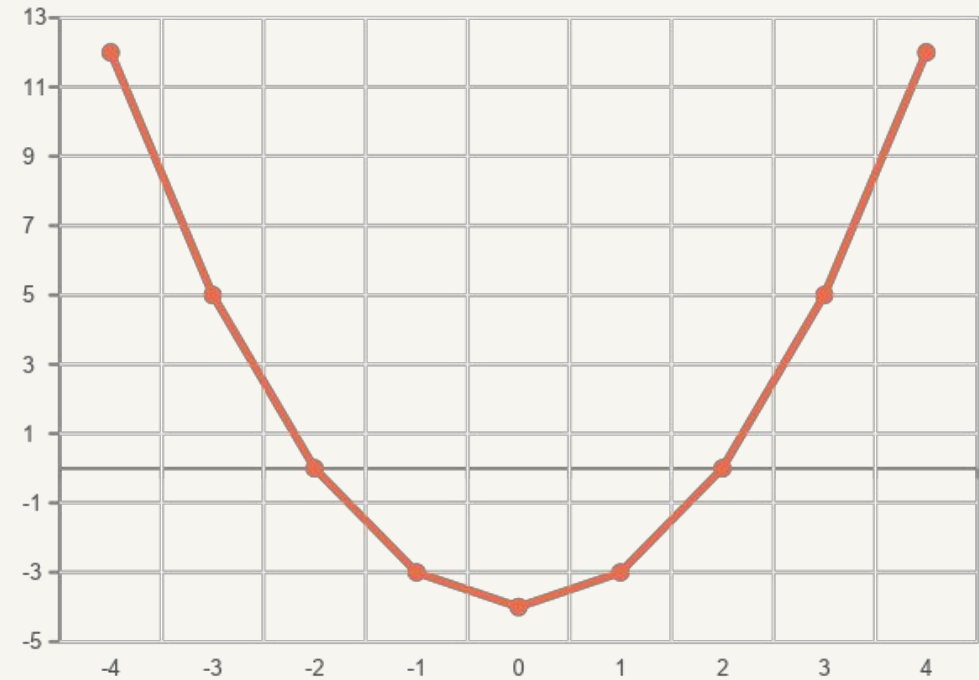
If $f(a)=0$, then a is a root

Example: $x^2 - 4 = 0$

Try $x = 2$: $2^2 - 4 = 0$

Try $x = -2$: $(-2)^2 - 4 = 0$

So the roots are -2 and $+2$.



Crosses x-axis → root

$x = -2$

$x = +2$

Two common types of equations

The same goal: solve $f(x)=0$, but the equation structure is different.

Algebraic

Built using powers of x .

Examples:

- $x^2 + 3x + 2 = 0$
- $x^3 - 7x + 5 = 0$

Small degrees may have formulas.

Transcendental

Includes functions like $\sin$, $\cos$, e^x , $\log x$.

Examples:

- $\sin x - x^2 = 0$
- $e^x - 5x = 0$

Usually solved approximately.

Numerical methods can solve both

Polynomial: the main algebraic form

A polynomial is a sum of powers of x with constant coefficients.

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n$$

**n = degree
(highest power)**

**$a_0, a_1, \dots$ are
constant coefficients**

**$a_0 \neq 0$
so degree is valid**

**a_n = last term
(constant)**

Examples by degree

Degree 2: $x^2 - 5x + 6$

Degree 3: $2x^3 - 4x^2 + 5x - 1$

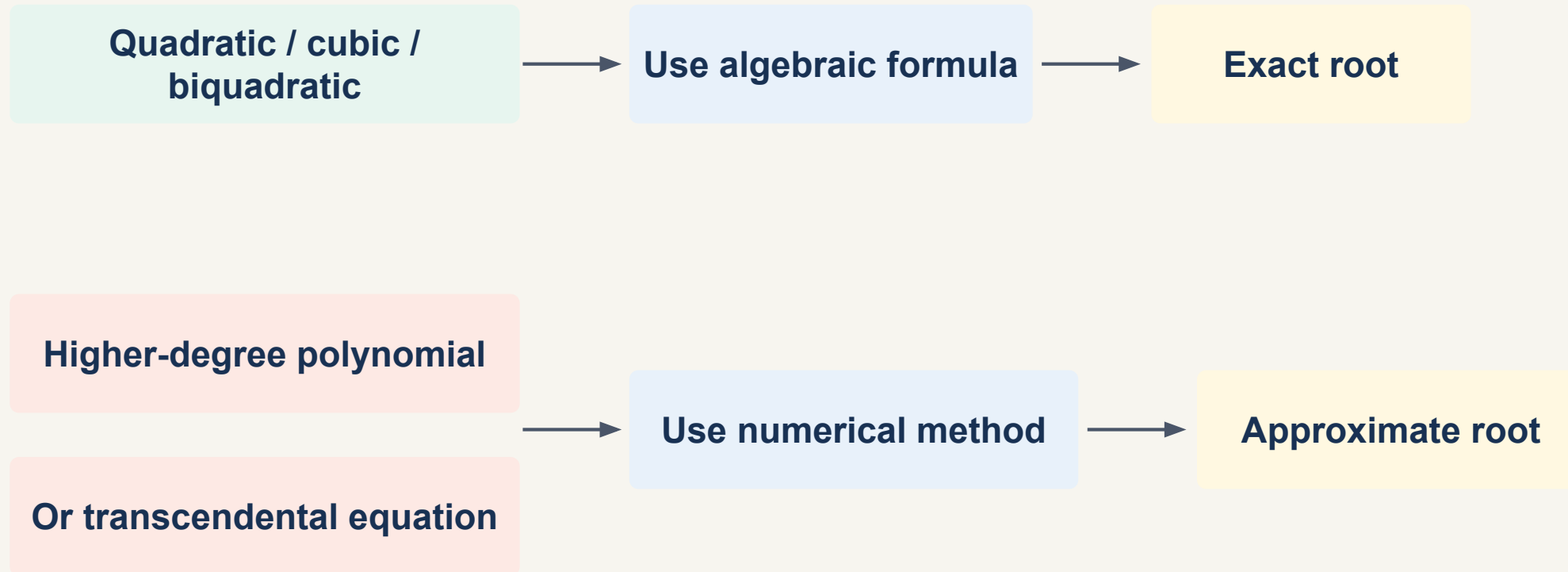
Degree 4: $x^4 + x^2 - 3$

Meaning

The degree tells us the maximum number of roots and helps choose the solving method.

Why do we need numerical methods?

Exact formulas are not always practical or available.



Examples of numerical methods: Bisection, Newton–Raphson, Secant.

Useful facts before solving

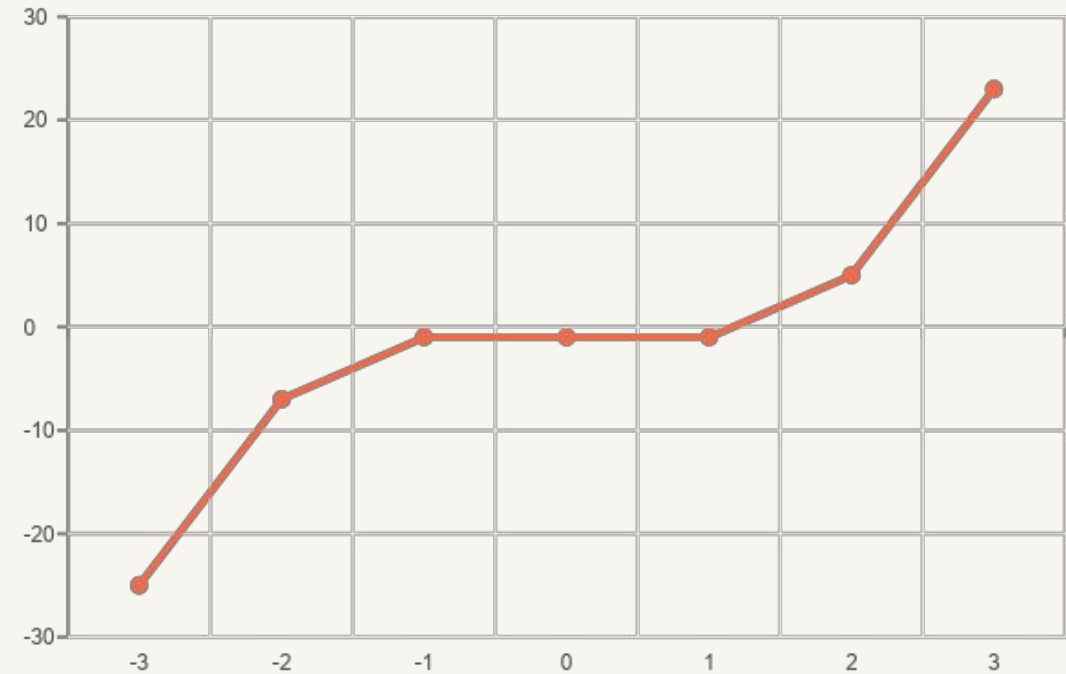
These facts help us understand how many roots may exist.

**Degree n polynomial
has exactly n roots**

Some roots can be real, and some can be complex.

Example: a degree-3 polynomial has 3 roots in total.

**Odd degree polynomial
has at least one real root**



Cubic curve must cross the x-axis at least once

Roots often appear in pairs

For polynomials with real/rational coefficients, some roots come together.

$$x^2 - 4 = 0 \rightarrow x = -2, +2$$

Even degree + negative constant term can give one positive and one negative real root.

$$x^2 + 4 = 0 \rightarrow x = +2i, -2i$$

If coefficients are real, imaginary roots occur as conjugate pairs.

$$x^2 - 2 = 0 \rightarrow x = +\sqrt{2}, -\sqrt{2}$$

If coefficients are rational, irrational roots also appear in related pairs.

Quick recap

Before applying a method, understand the equation first.

1. Identify equation type

Algebraic or transcendental?



2. Check degree and roots

How many roots may exist?



3. Use sign rules

Positive, negative, paired?

Then solve using Bisection, Newton–Raphson, or Secant Method

Main message: roots are the x-values where the graph touches or crosses $f(x)=0$.